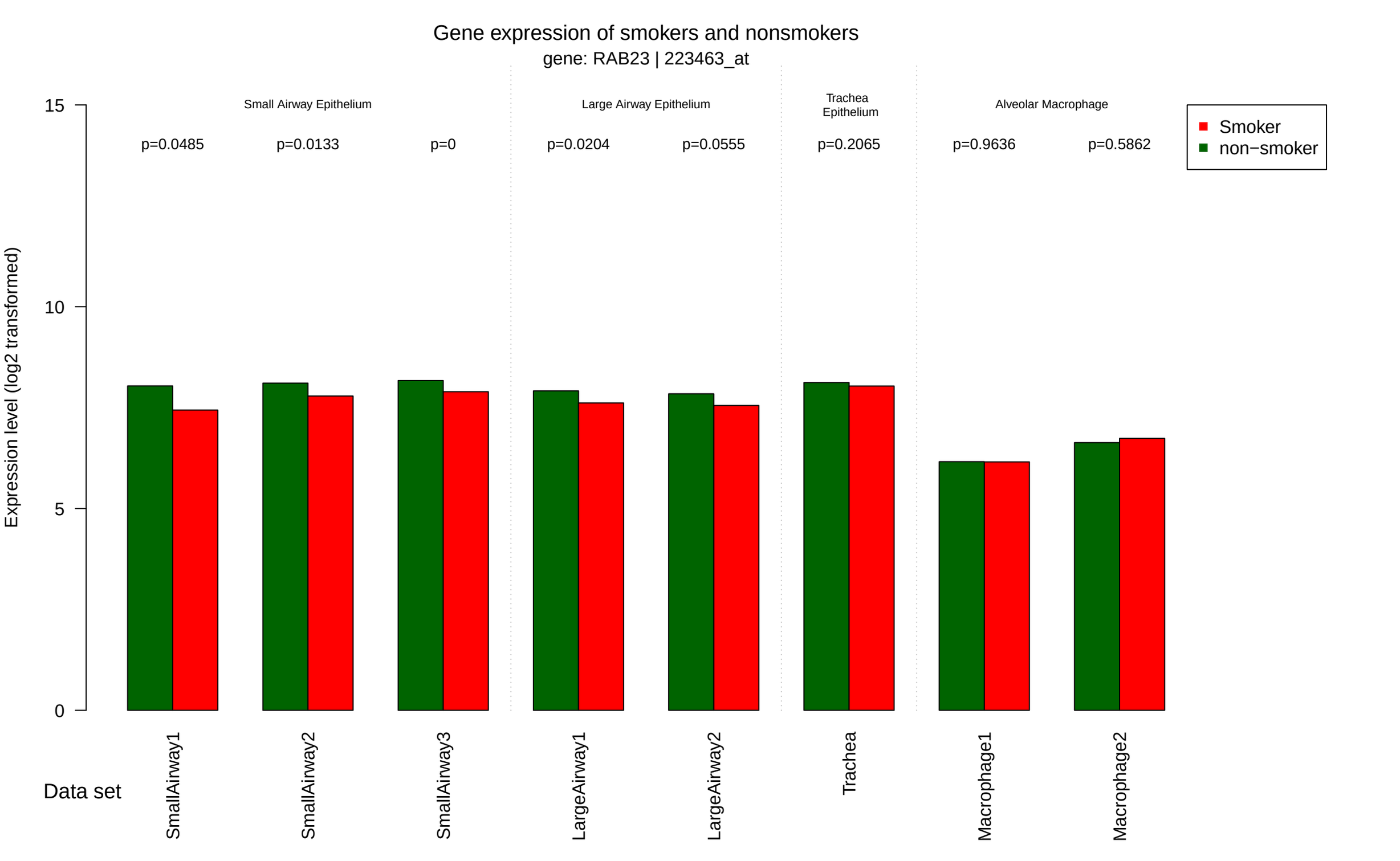
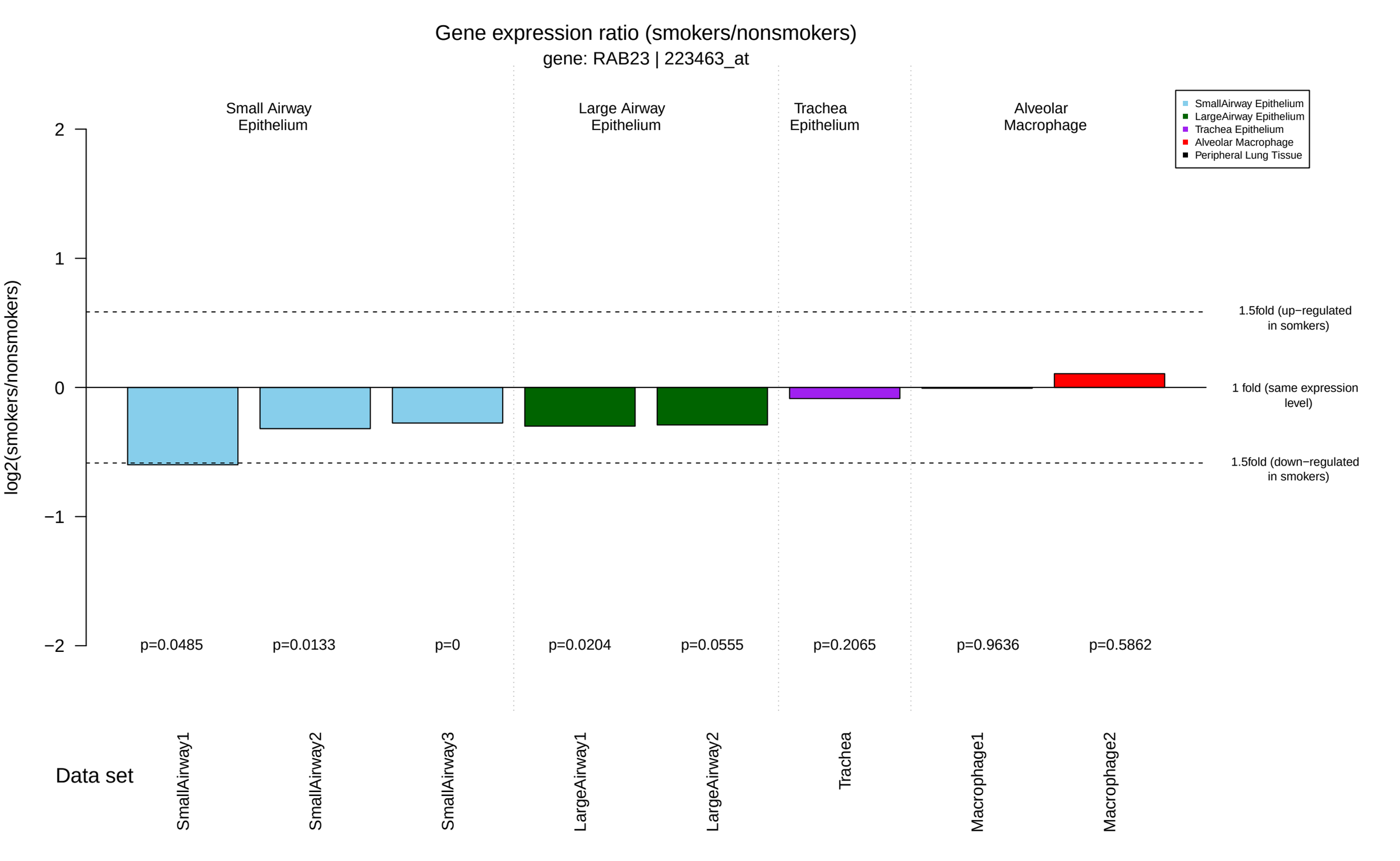
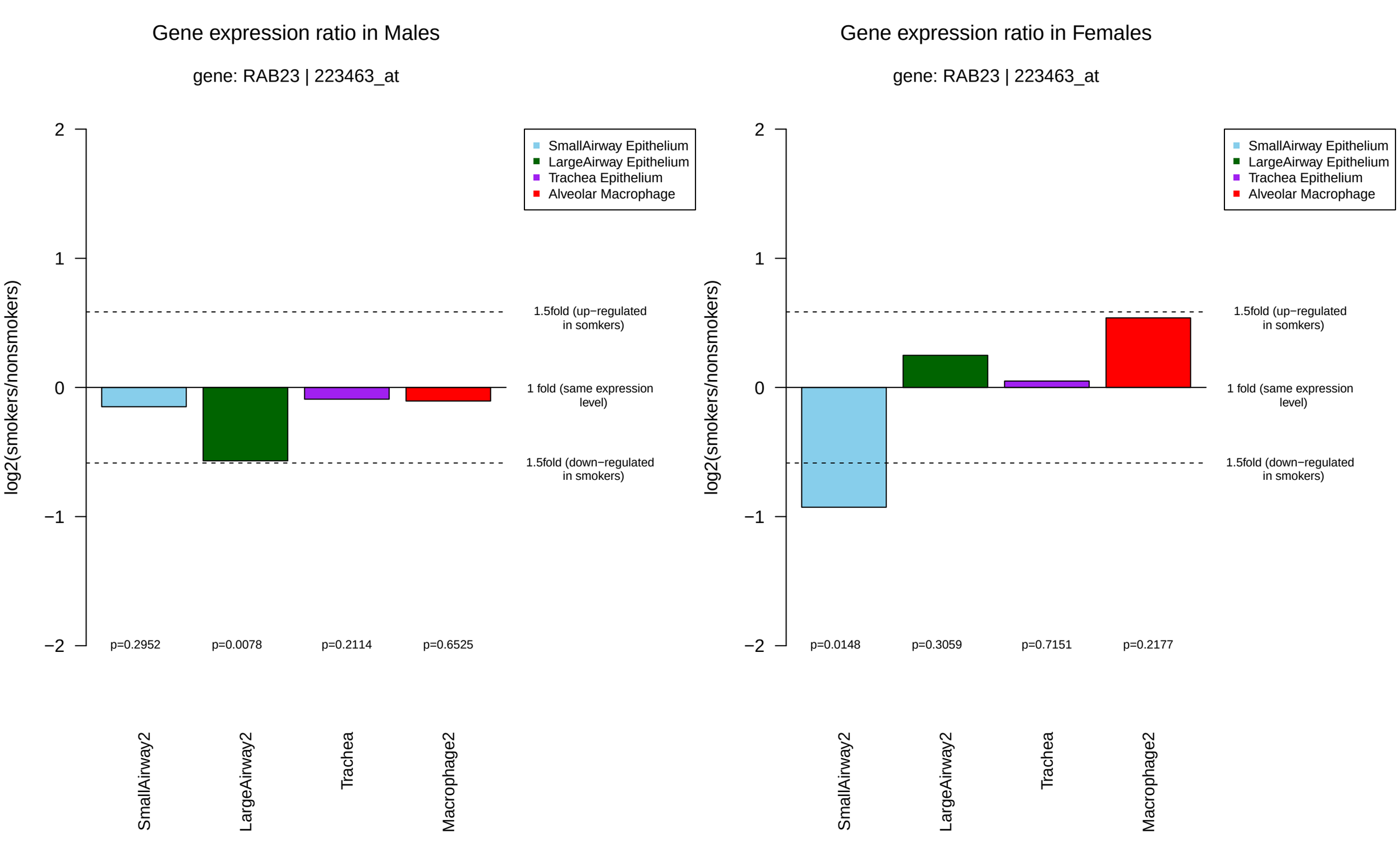
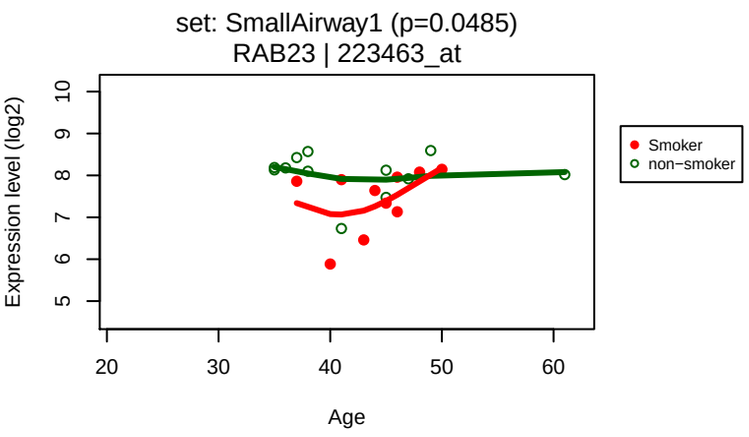
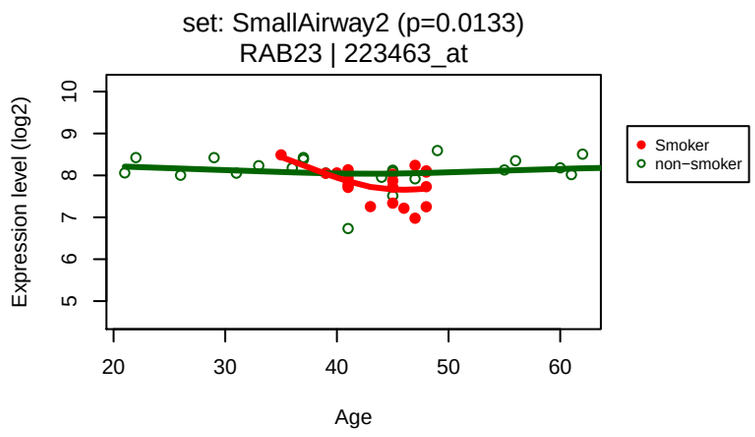
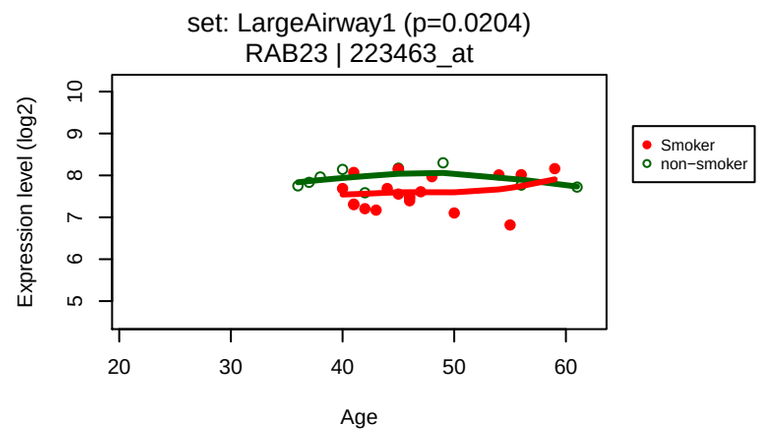

Gene expression of smokers and nonsmokers
gene: RAB23 | 223463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0485
p=0.0133
p=0
p=0.0204
p=0.0555
p=0.2065
p=0.9636
p=0.5862
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

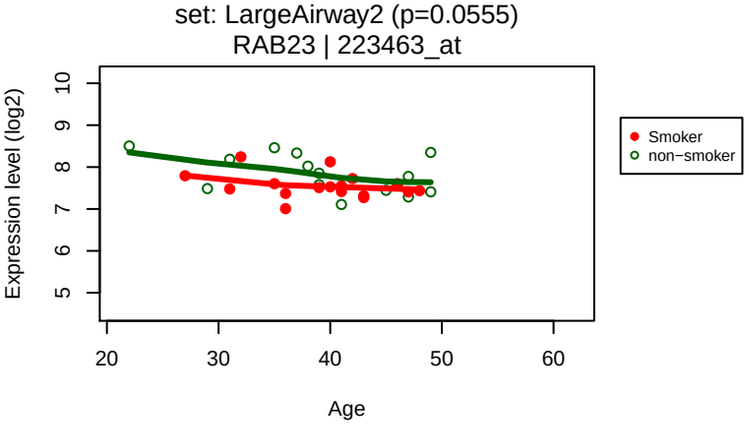

Gene expression ratio (smokers/nonsmokers)
gene: RAB23 | 223463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0485
p=0.0133
p=0
p=0.0204
p=0.0555
p=0.2065
p=0.9636
p=0.5862
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

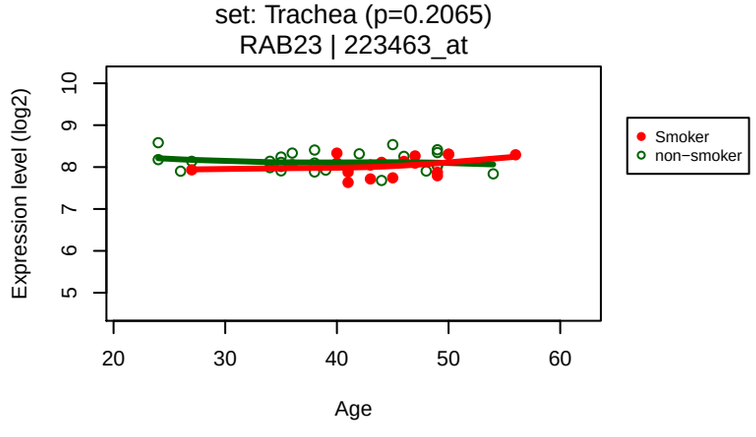

Gene expression ratio in Males
gene: RAB23 | 223463_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2952
p=0.0078
p=0.2114
p=0.6525
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB23 | 223463_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0148
p=0.3059
p=0.7151
p=0.2177
SmallAirway2
LargeAirway2
Trachea
Macrophage2

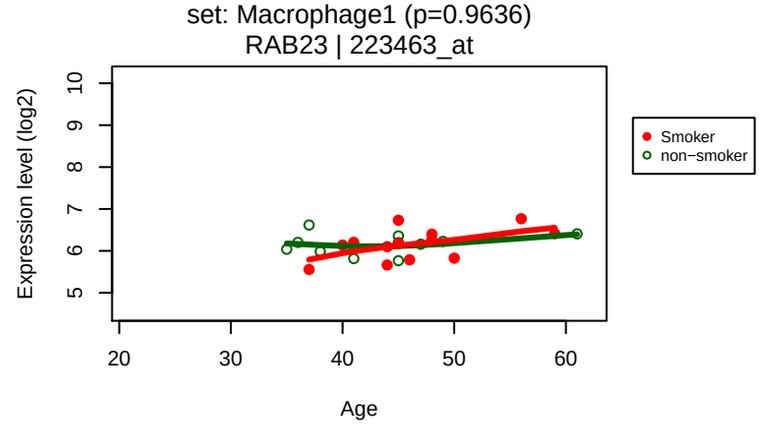

set: SmallAirway1 (p=0.0485)
RAB23 | 223463_at
Expression level (log2)
Age
Smoker
non−smoker

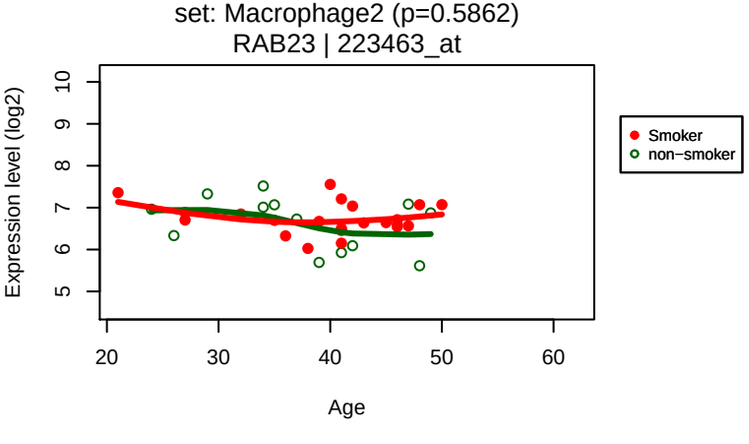

set: SmallAirway2 (p=0.0133)
RAB23 | 223463_at
Expression level (log2)
Age
Smoker
non−smoker

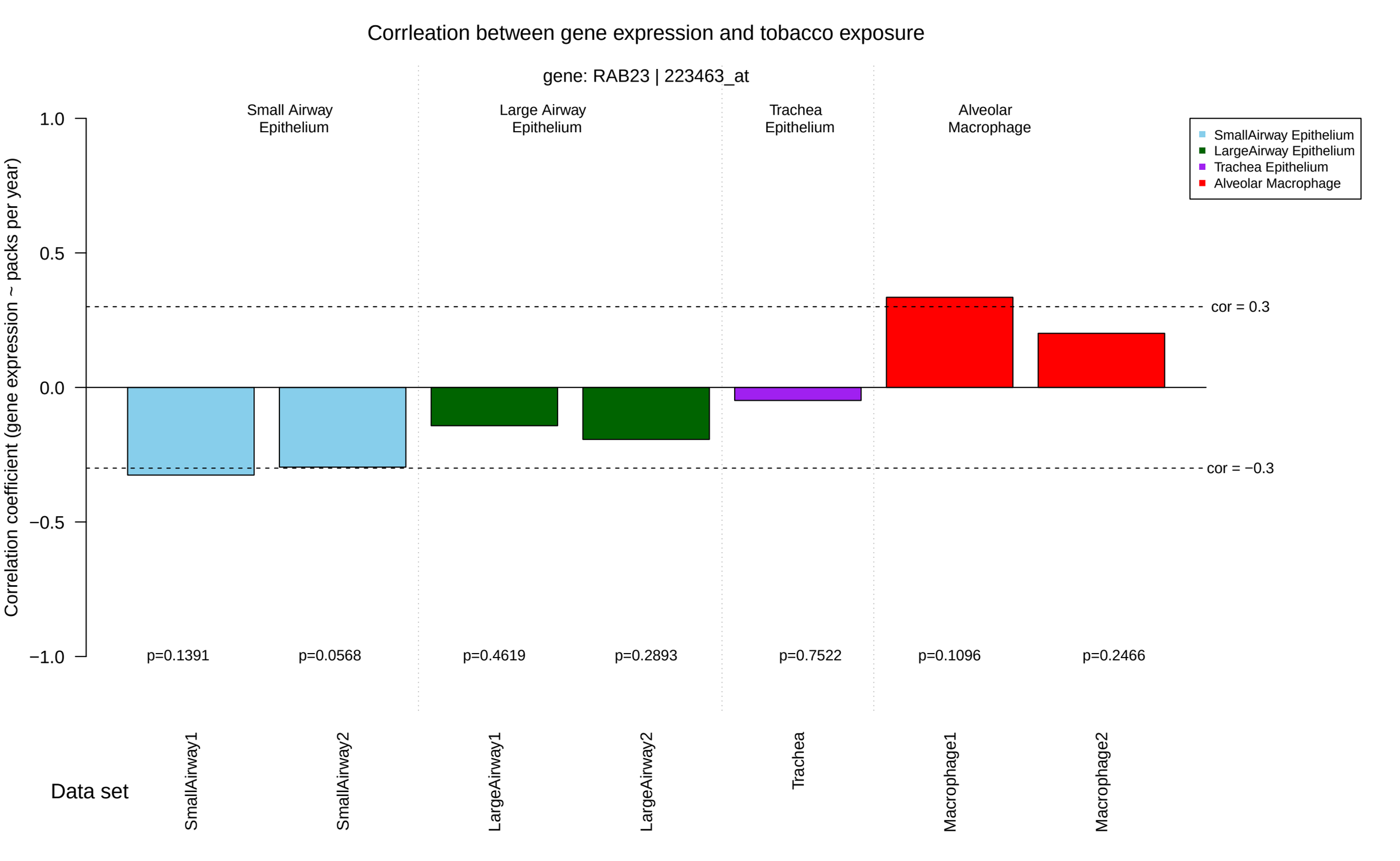

set: LargeAirway1 (p=0.0204)
RAB23 | 223463_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0555)
RAB23 | 223463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2065)
RAB23 | 223463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9636)
RAB23 | 223463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5862)
RAB23 | 223463_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB23 | 223463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1391
p=0.0568
p=0.4619
p=0.2893
p=0.7522
p=0.1096
p=0.2466
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2